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Getting a clue: creating student detectives and dragon slayers in your library

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Abstract

Purpose – The purpose of this paper is to describe how Utah Valley University Library staff created two games to orient students to the library and library services.

Design/methodology/approach – Library staff developed and marketed the Get a Clue game, which used clues placed throughout the building at the beginning of the Fall semester to orient new students as they solved a mystery. During the Spring semester, the library staff introduced library services through LibraryCraft, an online game where students used library resources to slay a dragon.

Findings – In post-game surveys, students found the games entertaining and informative. The results show students saw the orientations as a good use of their time and their comfort levels with library services increased.

Practical implications – As a means of engaging and informing students, games offer a new means of orienting students to the library and library services. The self-paced game approach allows students to learn valuable information with minimal impact on staff.

Originality/value – This paper offers practical information about using games as an academic library orientation. Assessment data support the effectiveness of games as an effective, asynchronous method of introducing students to a library facility and services. This information can be used by other libraries to create their own self-paced orientation games.

Keywords Gaming, Academic libraries, Orientation, Information literacy, Students, Learning, United States of America, Management games

Paper type Case study

Introduction

Utah Valley University (UVU) currently enrolls more than 32,000 students and is the second largest public four-year institution in Utah. After experiencing tremendous growth in recent years, the university remains student-centered with a commitment to open admissions for undergraduate students. With a history as a technical school and then community college, UVU gained university status in July 2008. Offering over 120 baccalaureate, associate, certificate, and diploma programs, UVU added three Master's degrees in education, nursing, and business and opened a new 190,000 ft^2 library facility to coincide with the change in status.

During the Spring 2009 semester, the UVU Library worked with a member of the behavioral sciences faculty and student researchers to conduct focus groups on student perception of the library and its services. In three focus group sessions, students identified the library services they used, as well as barriers to using the library. Students identified the size of the building and a lack of knowledge about library services as the most challenging barriers.

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At the time, the UVU Library did not offer a formal orientation to new students. The only introduction to the library came from the university's new student orientations, which include a campus tour with a stop in the library, and a booth at each event where library staff hand out brochures and library maps. Librarians also provide bibliographic instruction on request to freshman level courses such as Introductory Writing (ENGL 1010) and University Student Success (CLSS 1000).

Based on the focus group findings, student researchers identified a more thorough orientation as the most important way for the library to help new students effectively use the new building, as well as communicate the services available. The literature indicates that students recognize the value of library orientation and touring library facilities, as well as feel more comfortable using the facility after having an introduction (Kearns, 2010; Mosley, 1997).

The student researchers suggested offering a class before the Fall semester or additional library orientation. However, the library has not successfully offered open workshops in the past, and staff did not see this as a viable solution. UVU's commuter campus culture has made it difficult to advertise these workshops. Many students leave campus after their classes are over and do not return for extracurricular events and activities. With a record number of new students and a staff of 12 librarians and 20 paraprofessionals, the staff could not meet the demand of orientation sessions for thousands of incoming students. Based on experiences in other libraries, asynchronous learning experiences can be as effective as guided orientation (Mikkelsen and Davidson, 2011). As a result, rather than offering tours or workshops, members of the library's promotions committee determined that self-guided games could introduce students to the library and its services in an engaging manner.

This paper will describe the process of creating and launching two self-paced library orientation games, Get a Clue and LibraryCraft, and assess their effectiveness in achieving the library's goals. This paper will also identify and discuss the principal challenges of orientation games and strategies for overcoming those challenges.

Literature review

Librarians have found many reasons for students' lack of interest in library instruction: the perception that they already know the material, repetitive library instruction sessions, or the content being viewed as dull and tedious. Research strategies, methods, and mechanics do not interest them. Games may reach the current generation of students in ways other methods do not (Leach and Sugarman, 2005). Games can change students' perceptions of librarians by showing them as approachable and "in touch" (Doshi, 2006). The usual interactions of helping students at the reference desk or teaching instruction sessions only allows librarians to reach a small portion of the students who need assistance, while games can reach a broader audience as an unlimited number of students can work through a game at any time (Markey *et al.*, 2009). As Danforth (2011) writes, "There is a world of promise for libraries and other institutions supporting meaningful and valuable intellectual endeavors through well-designed gamifications [the inclusion of game-like activities or rewards outside of games]" (p. 84).

The current generation of new students, Millennials, present an additional challenge with a number of unique characteristics, especially in their learning styles and expectations. These students often prefer teamwork, experiential learning, and technology. They expect services to be open and available to them 24/7 in a variety of modes with quick response times and are more likely to use internet sources than visiting the library. With the vast amount of information available, the world of the Millennials differs from their predecessors (Frand, 2000). The rapid speed of changing information allows students to see an "altered time and space" with information being shared as events occur, causing the life of information to be much shorter, where past generations saw information last a lifetime (Frand, 2000, p. 17).

One of the defining characteristics of the current generation of students is the infiltration of gaming into the common culture. According to the Pew Research Center, 80 percent of teens aged 12-17 play five or more different game genres while 40 percent play eight or more different genres (Lenhart *et al.*, 2008a, b). The prevalence of gaming is quite striking when compared to older generations. Overall, 53 percent of adults 18 and over play video games, with 21 percent playing every day or almost every day. When looking specifically at younger adults ages 18-29, the number of gamers increases, with 81 percent reporting that they play games online and offline using a computer, cell phone, or other gaming device. By comparison, 60 percent of 30-49 year old adults report playing video games and only 40 percent of 50-64 year olds (Lenhart *et al.*, 2008).

With the pervasiveness of gaming by students entering higher education, attention has turned to gaming's educational foundations and its impact on student learning. Researchers have found that gaming has much in common with instructional theories such as problem-based learning or case-based studies (Landsberger, 2004; Oblinger, 2003; Broussard, 2010; Garris *et al.*, 2002). For researchers, the appeal of gaming comes from the impact on student thinking and learning patterns. Interactions with games can affect how students learn. Gee (2007) proposes that games help players think like scientists by causing them to "hypothesize, probe the world, get a reaction, reflect on the results, reprobe to get better results" (p. 216). In a similar example, Frand (2000) refers to the Millennials as the "Nintendo generation" since they learn by trial and error (p. 17). For students with this mindset, each loss is a learning experience that may lead to a deeper understanding (Frand, 2000). Games allow students to experience a more powerful form of learning. One of the most appealing aspects is the ability to make complex ideas more accessible to a variety of students (Squire and Jenkins, 2003). No longer simply memorizing facts, students can connect and manipulate information from outside the game and apply it to the game environment (Gee, 2003; Squire and Jenkins, 2003). When players use a simulation, they can learn to visualize complex systems and turn complex text or numbers into a format that is easier to comprehend (Oblinger, 2003).

To accomplish educational objectives effectively, game-based learning must be based on sound pedagogy and reflect learning needs in a "real educational setting" with clear rules and goals, as well as being fun (Hong *et al.*, 2009, p. 431). The training simulations and visualizations with interactivity produce greater motivation, higher completion rates, wider participation, better collaborative learning, and quicker learning times. However, educational games cannot be deemed successful if not closely aligned with the learning outcomes, even when the game produces greater motivation (de Freitas, 2006). Also of note is the need to keep a game's focus on a small set of goals. To be successful, games should target specific objectives and not attempt to be comprehensive (Markey *et al.*, 2008).

Games in higher education have been documented in a number of fields, including medicine, education, and literacy (Kanthan and Senger, 2011; Kablan, 2010; Kambouri *et al.*, 2006). Using games in libraries has proved quite popular. Examples of games include treasure hunts, mystery tours, puzzle hunts, book hunts, web-based board games, and games based on popular television shows such as Jeopardy! and The Amazing Race (Boule, 2009; Marcus and Beck, 2003; Kashbohm *et al.*, 2006; Gallegos and Allgood, 2007; Thompson *et al.*, 2008; Markey *et al.*, 2009; Markey *et al.*, 2008; Walker, 2008; Broussard, 2010; de Freitas, 2006). Depending on the skill level of the staff and the time available, more elaborate games may be possible. At the University of Calgary library, attempts are being made to modify the commercial video game Half Life 2 into an information literacy game (Clyde and Thomas, 2008).

Although educational goals are critical, maintaining the basic principles of gaming also improves success. The regime of competence principle should be incorporated into games, meaning games should be "doable" but not too easy, pushing the student to gain mastery at one level while forcing them to adapt and evolve (Garris *et al.*, 2002; Gee, 2007). Games should include help, cues/prompts, and other activities to support the learner (Garris *et al.*, 2002). The overall design also impacts how well players accept the game. Students can tolerate "low-budget" games, but are frustrated by bad design such as "violating the norms in terms of controls, pacing, interface, modes of interaction and so on" (Landsberger, 2004, p. 6). Danforth (2011) warns of adding game elements to "nongame settings" without regard to how games actually function:

> Game mechanics are being tacked on to practically everything these days, almost as an afterthought. The resulting experience, distastefully shallow and frankly venial, threatens to overshadow the deeper value of well-designed, substantive games that entertain through genuine engagement, demand creative thinking, and lend insight into the human condition through vivid storytelling (Danforth, 2011, p. 84).

The successful use of games in education hinges on using a known setting, situation, experience, or applying a familiar game in a new way. Adapting or borrowing from a known game reduces the amount of explanation required (Leach and Sugarman, 2005). Learners are more willing to engage with a game that is familiar to them. Offering students a point of reference within a new game fosters confidence (Kambouri *et al.*, 2006).

Although promising, gaming by itself is not the sole answer for reaching Millennials. Gaming should coordinate technology use with the content, students' learning style, and instructor's style (Frand, 2000). In addition, an effective conflict and story will turn an activity into a game by stimulating emotions in the players and making an activity fun (Broussard, 2010). Designers face a challenge to adapt game features in a way that sustains interest while maintaining the enjoyment level.

Get a Clue

Objectives of Get a Clue

The UVU Library's Promotions Committee – composed of librarians and staff members from several departments in the library – discussed the student researchers' orientation suggestion and explored ways to implement it. The committee determined a formal class would not attract students and attendance would be minimal. After ruling out a formal orientation, the committee decided to use a game to introduce new students to the library. To reduce the impact on an already busy staff, the committee envisioned a self-paced experience where students would move through the building, learning more about library services and resources as they went. The committee identified major objectives the game would need to achieve. The primary objective was to introduce new students to the physical layout of the building. The new 190,000 ft^2, five-story library overwhelmed new students, many of them from small, rural areas with minimal library resources. Other game objectives included introducing freshman students to the basic services of an academic library and registering them in the integrated library system (ILS) to expedite the check out process.

As the literature shows, self-guided tours can be highly effective, and an educational game seemed the perfect solution for student engagement (Marcus and Beck, 2003). Research has also shown that using a known game provides a valuable point of reference, the children's game Clue seemed a natural fit as the theme of the game (Kambouri *et al.*, 2006). Games used in other libraries – murder mysteries, races, scavenger hunts – were deemed too complicated or too time-consuming to set up. In Clue, players are asked to visit locations on the board and deduce the specifics of the crime by process of elimination. For the orientation game, the committee decided to follow a similar premise. Students would eliminate suspects and library resources as they worked their way through the building. Using a familiar game would also simplify matters for the members of the promotions committee. Prior to creating Get a Clue, none of the committee members had experience creating an information literacy-based or orientation game.

Planning and gameplay

Get a Clue's planning and design phase took less than two months. With the goals and premise set, the committee worked out the details of the game and began the initial design work on the game piece. One side of the 8.5″ × 11″ game piece (see Figure 1) showed floor maps. The opposite side included the rules, descriptions of important locations around the library, and an entry form for the student's solution to the mystery. The location descriptions included key facts about library services and collections, providing necessary background information. The game was designed to take approximately 30 minutes to complete. If the game took too long to complete, students would lose interest; if the game was over too quickly, the goal of introducing students to the building layout could not be met.

After completing the initial script for Get a Clue, committee members reviewed it for clarity Student workers tested the game to determine what changes were needed before completing the artwork for the clue posters. Each clue was printed on 12″ × 18″ paper, mounted on matte board, and consisted of the clue text and images of staff members portraying characters in the game (see Figure 2). The clues were designed to help students eliminate a person, weapon, and place, as well as lead them to the next location.

An opening video podcast introduced the characters of the game and established the scenario. Using the clues, students traveled throughout the library. These clues were built around a central story, a mystery, and followed a detective through the steps of solving a crime. To make the game more entertaining, the inept detective followed in the tradition of Inspector Clouseau from the Pink Panther movies An excerpt from Get a Clue text is shown below:

> Schmolmes: To the moveable shelving! Bring the fingerprinting kit!
> SCHMOLMES'S NOTE: Locate moveable shelving in the library. Forgot to ask where it was before dashing off. Forget own head next.

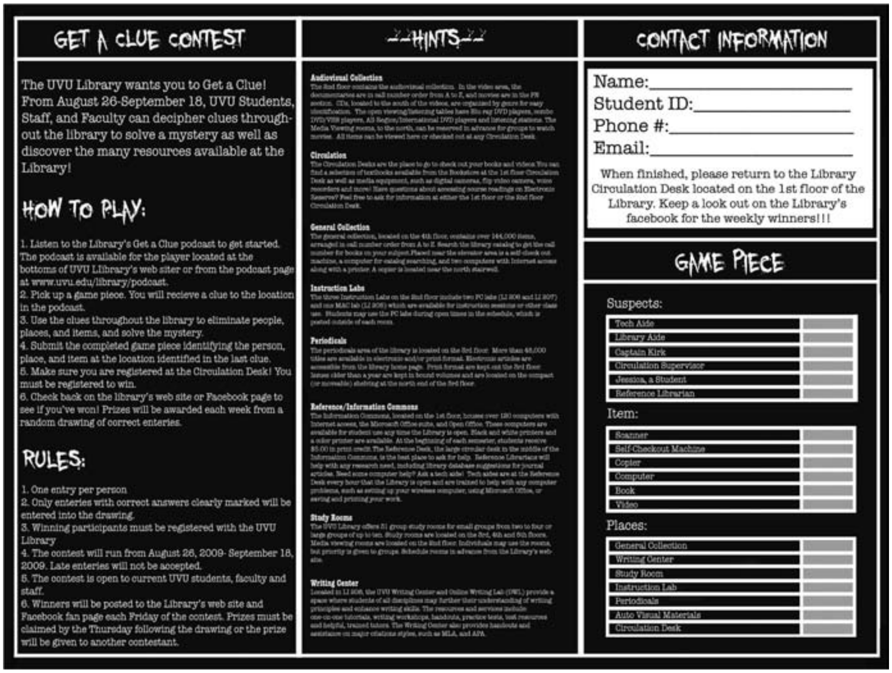

Figure 1.
Get a Clue game piece, which includes the game instructions and rules, descriptions of library locations, and an entry form

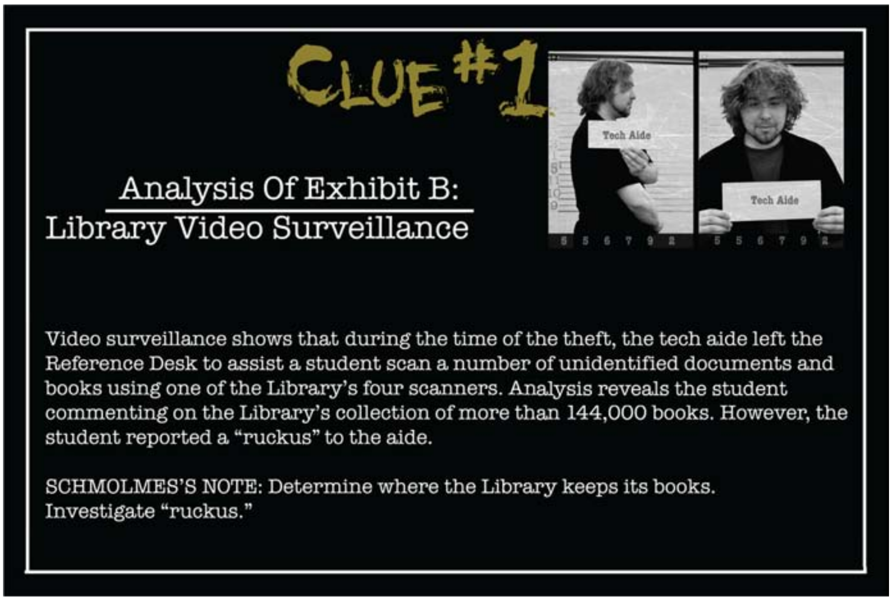

Figure 2.
Sample clue from Get a Clue. Each of the clues provides information to eliminate a suspect and points the student toward the next clue's location

The game launched with the beginning of Fall 2009 and ran for the first three weeks of the semester. A prize drawing was offered to entice students to play the game. The inexpensive prizes included gift certificates for local businesses and office supplies from the campus bookstore. To be eligible, students needed to register with the library and have the correct person, location, and weapon identified on their entry form. The library held three drawings during the course of the game and winners were notified by phone and through postings on the library's Facebook page.

Issues and changes for 2010 version

The subject librarian introduced the game to the English faculty shortly before the Fall 2009 semester. The faculty received the idea with a great deal of interest, and many required their students to complete the game or gave extra credit to students completing the game. As shown in other studies, gaming tied to course credit will improve success (Markey *et al.*, 2009). However, this presented a problem the committee did not anticipate. For students to enter the prize drawing, the library needed the entry form submitted, leaving students with nothing to give to their instructor as proof of completion.

Of the 311 entries submitted, only 172 correctly identified the game's villain (55 percent of the total). Although the number of correct answers was much lower than desired, the committee decided the game was more successful at achieving the more important goal of introducing students to the building. The game continued the next year, with modifications to improve the number of correct responses. Support from the faculty, as well as the participation rate, made it a worthwhile endeavor.

The library revised the game for the Fall 2010 semester. Students no longer needed to eliminate items to determine how and where the Clue-inspired crime had been committed. Instead, students followed the clues throughout the building and recorded the various locations. The committee eliminated the video introduction since students did not watch it. Students no longer needed to register, as this added a burden on the busy circulation staff. To provide students with proof they had completed the game, the entry forms were no longer part of the game piece. Once students finished playing, a librarian at the reference desk stamped the game piece, and students could submit this to their instructors. Students submitted a separate entry form for the prize drawing. With these changes, students completed the game quickly with minimal staff intervention.

Results

Upon completion of the Fall 2010 version of the game, 338 students responded to a short survey to assess how well the new version of the Get a Clue game met its intended goals. Based on the responses, 88.5 percent completed the game as part of a class assignment or for extra credit. The results indicated the game reached students who had minimal exposure to the library (the target audience), with 63 percent having visited the library fewer than five times (see Table I).

Table I.
Get a Clue survey results. Question 3 of the post-game survey showed that students found the game easy to complete

How easy was it to complete Get a Clue?	Response percentage	Response count
Very easy	19.6	67
Easy	72.1	246
Difficult	10.9	37
Very difficult	0.3	1

During the first run of Get a Clue, the promotions committee observed that confusion over the game's rules and requirements prevented students from achieving the more important goals of navigating the building and learning about the library's services. This time, 91.7 percent of students rated the game as very easy or easy, indicating that the revisions successfully alleviated students' confusion during the first version of the game.

Comments from students were very encouraging and enlightening, with most coming in the form of suggestions for improvement:

> Get stamped at every clue station.
>
> Make #5 easier to find.
>
> Make the game piece less obtrusive (less obvious).
>
> Make it harder. It was tons of fun.
>
> Nothing. It was an awesome fun experience.

As an orientation to the library, increasing the students' comfort level with the location of services and resources in the building was a major objective. Based on the responses (see Table II), this objective was accomplished with nearly 90 percent of students reporting they were more comfortable navigating their way around the library after completing Get a Clue. In addition, 92 percent of the students responding described the game as a good use of their time, and 92.3 percent would recommend the game to other students.

LibraryCraft

Objectives of LibraryCraft

LibraryCraft's objectives were somewhat different than the goals of Get a Clue. Instead of guiding students through the physical library, LibraryCraft guided students around the library's website. Before writing the game script and rules, the promotions committee identified essential library skills and under-advertised library services to emphasize. While students would not receive in-depth instruction, the committee intended for the game to provide an introduction to the library's online resources. In that way, LibraryCraft was designed to be a promotional tool that would educate incoming freshmen about the online services of an academic library and basic strategies for using those services. The planning and design phase of LibraryCraft was longer, lasting nearly three and a half months, because of the number of high-quality graphics needed for the game.

Planning and gameplay

To capitalize on the popularity of online games like World of Warcraft, LibraryCraft sent students on a medieval quest. The game used a simple, light-hearted story with

Table II. Get a Clue survey results: Question 4 of the post-game survey showed that students were more comfortable navigating the library after playing the game

After completing Get a Clue, how comfortable do you feel navigating your way around the library?	Response percentage	Response count
More comfortable	89.6	305
No change	10.3	35
Less comfortable	0	0

the student as the protagonist. Students chose one of two avatars to represent themselves for the duration of the game and made their way from task to task, visiting different parts of the library's website as they progressed. Using feedback from student library workers, the committee worked to make the game as entertaining as possible, while still achieving its educational goals.

A member of the promotions committee wrote the script and requested feedback from librarians and students, using experience gained from writing the script for Get a Clue. Tasks were created for students to complete as they worked through the game. Unlike Get a Clue, where students had to navigate their way through the library based on mystery clues, the tasks in LibraryCraft (see Figure 3) involved looking up books and articles or answering questions about services such as interlibrary loan and media equipment check out. Each correct answer would "unlock" a new chapter. Pop-up windows provided help at each task, offering strategies for searching or hints for finding information about various library services. In addition, librarians at the reference desk used a cheat sheet to help students who were stuck on a particular chapter.

The game awarded points for correct answers to the tasks, but students were allowed to try as many times as needed to progress to the next chapter. The game's tasks utilized simple JavaScript functions to request, receive, and accept user data. The JavaScript was made as robust as possible to accept variations of the correct answer, i.e. answers written in all capitals or all lower case. The committee member tasked with creating the web elements had some self-taught expertise in creating HTML documents and JavaScript coding. Tasks grew slightly more difficult over the course of the game to challenge and motivate students, as suggested by the literature on educational gaming (Garris *et al.*, 2002). The first tasks were simple, asking students to locate a specific book in the library catalog. By the end of the game, however, students searched through the library's web pages to find important information about services and decide on the best answer to complete the tasks. The literature reports that tying

Figure 3.
Sample Task from LibraryCraft. This task from LibraryCraft asks students to find a specific e-book in the library's collections

in-game tasks to "personal competencies" that are real-world skills will mean more to students than using challenging but obscure or irrelevant goals (Garris *et al.*, 2002).

Artwork for the game (see Figure 4) was modeled on popular computer games, complete with an inventory in the lower right of the screen showing the students' progress. This part of the design phase was the most time-intensive as the library staff had only one member with the requisite skills in graphic design and Photoshop. However, comments received from students who completed the end-of-game survey indicated that the time spent was more than worthwhile:

> The graphic pictures were amazing!
>
> The fact that it was a well thought out game with awesome graphics really appealed to me.
>
> I thought that LibraryCraft felt like a real game; I wasn't expecting the graphics to look like real pictures and the artwork made searching for the answers that much more enjoyable.

To attract interest, the library posted advertisements for the game on the library's homepage and the digital signage throughout the building. The subject librarian for the English department advertised directly to faculty who had incorporated Get a Clue into their classes during the previous Fall semester. As with Get a Clue, many students in lower-level English courses received class credit or extra credit for completing the game. The first run of LibraryCraft launched in February 2010 and ran for three weeks. Students who completed the game and the end-of-game survey were eligible for a prize drawing. Upon completion, students could print a certificate as proof of completion. Like Get a Clue, LibraryCraft took approximately 30 minutes to complete.

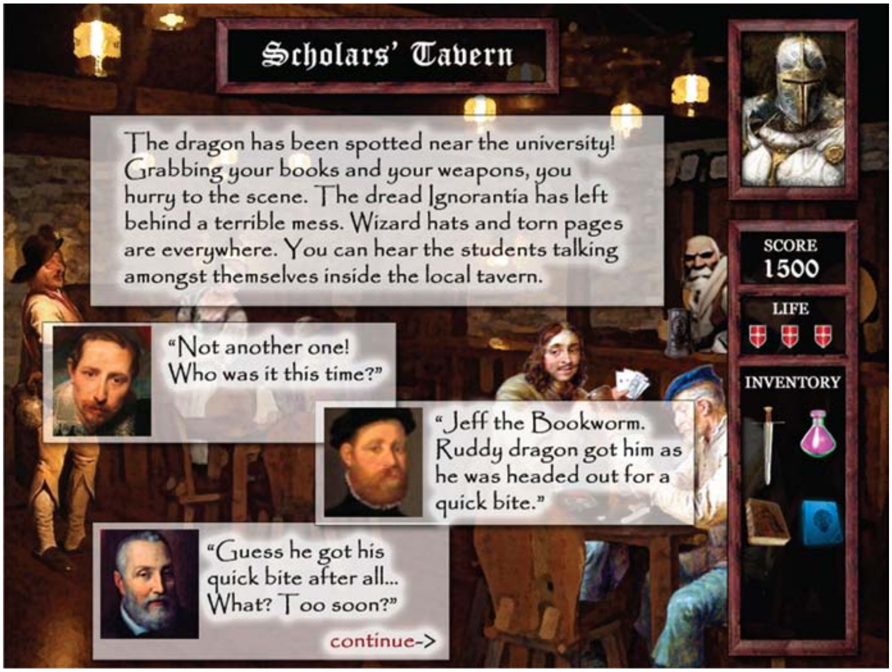

Figure 4.
Sample artwork from LibraryCraft. This passage from LibraryCraft shows the detail and quality of the game's artwork

Results

The Spring 2010 version of LibraryCraft concluded with a short, mandatory survey (see Table III). The first statement determined the success of LibraryCraft in reaching its educational goals. Of the 52 students who responded to this question, 84.5 percent agreed or strongly agreed that LibraryCraft taught them a new library skill. Students commented that the game helped them learn more about the library resources and performing research and – perhaps most importantly – made learning how to use those resources fun.

The survey also asked students what they disliked most about the game. While a few comments to this question were complimentary (11 percent) or neutral (37 percent), many responses (56 percent) reported difficulties using the game, confusion about the tasks, or indicated that the game was too long. However, 94 percent of respondents agreed or strongly agreed that the game was easy to follow. This feedback indicated that although some students struggled with the tasks, they were still able to complete the game. The disparity between the responses to these two survey questions suggests that LibraryCraft possesses the ideal level of difficulty: not too hard but not too easy (Garris *et al.*, 2002).

Comments on what the students enjoyed about the game were very encouraging for future editions of the game:

> It was a bit cheesy; But a great tool to learn how to use some of the things that are very useful with the library. Especially how it makes you look for certain things, and not just have the answer right there.

> I liked how they made it into a story. It made it fun and really enjoyable.

> It was pretty straightforward, helpful, in terms of learning the library's diversity, and very fun to complete.

> Creative way to teach a rather stale subject and made it entertaining as well.

> While it was giving information that can help me I found the story interesting. My daughter wanted to know what game I was playing? I told her it was an assignment for English. She said, "You play games for English? Cool. I want to play".

When asked if they would recommend LibraryCraft to their friends, 87 percent of respondents either agreed or strongly agreed that they would. While fewer students participated in LibraryCraft than Get a Clue, the survey results show that, overall, the game met its objectives and that most students enjoyed the experience.

The end-of-game survey comments were used to modify the game for its second run during the Spring of 2011. Based on student feedback, the promotions committee clarified and emphasized task instructions. The task interface was revamped with the instructions appearing on the left side of the screen in an HTML frame with a larger pane on the right

Table III. LibraryCraft survey responses. The first question on the post-game survey asked how many students credited LibraryCraft with helping them to learn a new skill

	Strongly agree		Agree		Disagree		Strongly disagree	
	n	Percent	*n*	Percent	*n*	Percent	*n*	Percent
LibraryCraft helped me learn a library skill I didn't already have	36.5	19	48	25	11.5	11	3.8	2

to navigate the library's databases and web pages on the right. The JavaScript elements of the tasks were thoroughly debugged and tested in a variety of browsers on several different operating systems to eliminate any lingering technical issues.

For several reasons, LibraryCraft was not as popular as Get a Clue. The library had direct contact with new students and heavily advertised Get a Clue during new student orientations, leading to higher participation. Librarians also spoke directly to a larger number of faculty, including adjunct faculty who taught lower division English and college success courses when introducing Get a Clue. Because Get a Clue was easier to set up, the library launched the game when the semester started. LibraryCraft was more complex and launched several weeks into the semester, past its intended launch date and during a busier part of the semester. The library's promotions committee learned from these mistakes and advertised the game much more effectively before its second run in Spring 2011.

Discussion

Overall, the library promotions committee determined these orientation games to be worthwhile endeavors. Both games achieve their educational goals with minimal impact on a small and busy staff, while still providing an entertaining experience for the students. Of students who played Get a Clue and responded to the end of game survey, 90 percent reported that they were more comfortable navigating the library as a result of the game. As for LibraryCraft, 84.5 percent of students who played and responded to the survey agreed or strongly agreed that the game taught them a new skill and commented that they had learned more about the library's resources. In both surveys, the majority of students reported that the games were a worthwhile use of their time and they had fun playing.

Since the first versions of Get a Clue and LibraryCraft launched in Fall 2009 and Spring 2010, respectively, the library has revised the games to eliminate the problems of the first versions. Several of the stumbling blocks of both games were due to complexity. Although the committee thought Clue was a well-known game, librarians at the reference desk quickly realized that many of the students playing were not familiar with the premise of eliminating clues to determine the answers. Get a Clue had too many different requirements for students to complete the game. The second iteration removed the registration requirement and Clue-inspired elimination elements. In the case of LibraryCraft, the design and programming of the game taxed the abilities of the staff members to finish it on time in addition to their regular duties. Providing better instructions for the tasks and simplifying the structure of the game not only helped more students finish, but also helped the library staff keep the game up to date as e-books used as clues disappeared from online collections and URLs changed.

Reaching the right level of difficulty was another challenge. Because UVU is an open admissions institution that also offers Master's degrees, student ability and familiarity with library resources range widely. While the members of the promotions committee did not want the games to be too easy and bore students, the games needed to accommodate students with low reading skills. In the first version of Get a Clue, students had difficulty with certain words in the clues, such as "ruckus", and were unable to get past them to decode the next location. In developing orientation games, various skill levels of the students must be kept firmly in mind and accommodated as much as possible without sacrificing too much of the motivation through challenge factor.

Additionally, achieving the educational goals while keeping the experience enjoyable for students was one of the most challenging aspects of creating these games. During the development process for both games, a determined effort was made to get feedback from the library's student workers on the artwork, the game story lines, the wording of clues and tasks, and the overall experience of playing the game. This feedback helped the committee members maintain an entertainment level appropriate for the Millennial generation and avoid design problems, such as those identified by Kurt Squire (Landsberger, 2004). Making sure that the game met its learning objectives was the primary goal. Indeed, this should be the primary objective of any group seeking to create a self-guided orientation game, but without the entertainment factor, students would not have been as motivated to complete the games.

Advertising is a key factor in the success of the games. Communicating with a large student body on a commuter campus is difficult. In anticipation of the second runs of Get a Clue and LibraryCraft, the library posted "teaser" ads to its website and to the digital signage posted throughout the building. The ads changed weekly leading up to the start dates. Library staff also advertised Get a Clue during new student orientations held throughout the summer. Prize drawings helped to draw students to both games during their first runs. Prizes were donated by local businesses and UVU organizations. Additionally, $75 was allocated to purchase more prizes.

Faculty buy-in remains a vital ingredient for success with orientation games. Without faculty support, these games would have suffered from lack of participation. Many of the faculty approached by the promotions committee required or offered extra credit points for participation in the game, overcoming this particular hurdle. Even when the game is over, it is important to keep in touch with faculty. In the case of Get a Clue, the library promotions committee discovered that the game was so popular that some faculty had put the game on their syllabi for later semesters, even when it was not available. When Get a Clue ran for a second time in Fall 2010, almost 400 students played. Seventy-eight students finished the second run of LibraryCraft. This is slightly lower than the completion rate for the first run, since the library did not offer prize drawings to entice students to play.

Conclusion

Get a Clue and LibraryCraft represent an innovative method for introducing new students to the library and its services through educational games. Because of the large numbers of incoming students, UVU cannot provide them all a tour or an orientation workshop. The cost to the library to develop these games was minimal since the promotions committee used the staff's talents and skills to create the game scripts and artwork. However, if the necessary skills are not available in-house, library staff could collaborate with other departments on campus. For libraries that have staff with basic HTML, JavaScript, and graphic design skills, games like LibraryCraft are an easy possibility. Get a Clue required nothing more than a good story and artwork – no special skills were needed to create it. The games allowed students to explore the library and discover the collections and services at their own pace. While the promotions committee anticipated problems, each run of Get a Clue and LibraryCraft has been a learning experience. The games have grown popular with both students and faculty, based on feedback received from both groups.

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